

Problem 1. Find the derivative

$$\frac{d}{dx} \left\{ \int_{\sqrt{x}}^{\cos x} \frac{\sqrt{t^3+1}}{t+7} dt \right\}.$$

Sln.

$$\frac{d}{dx} \left\{ \int_{f(x)}^{h(x)} g(t) dt \right\} = g(h(x)) \cdot h'(x) - g(f(x)) \cdot f'(x).$$

$$\begin{aligned} \int_{f(x)}^{h(x)} g(t) dt &= \int_{f(x)}^0 g(t) dt + \int_0^{h(x)} g(t) dt \\ &= \int_0^{h(x)} g(t) dt - \int_0^{f(x)} g(t) dt. \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} \left\{ \int_{\sqrt{x}}^{\cos x} \frac{\sqrt{t^3+1}}{t+7} dt \right\} &= \frac{\sqrt{\cos^3 x + 1}}{\cos x + 7} \cdot \frac{d}{dx}(\cos x) - \frac{\sqrt{x^{3/2} + 1}}{\sqrt{x} + 7} \cdot \frac{d}{dx}(\sqrt{x}). \\ &= -\frac{\sin x \sqrt{\cos^3 x + 1}}{\cos x + 7} - \frac{\sqrt{x^{3/2} + 1}}{2\sqrt{x}(\sqrt{x} + 7)}. \end{aligned}$$

Problem 2. A particle moves along a straight line so that the velocity at time t is $v(t) = 2t - 6$. Find the total distance traveled during the time period $0 \leq t \leq 4$.

Sln. Notice that $v(t) = 2t - 6$

$$v(t) \leq 0 \quad \text{for} \quad 0 \leq t \leq 3$$

$$v(t) \geq 0 \quad \text{for} \quad 3 \leq t \leq 4.$$

$$\mathcal{T}_d = \int_0^4 |v(t)| dt = \int_0^3 |v(t)| dt + \int_3^4 |v(t)| dt.$$

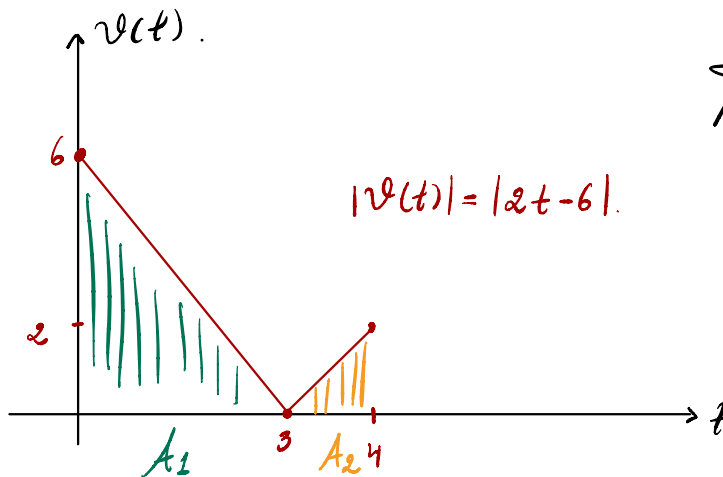
$$= \int_0^3 -v(t) dt + \int_3^4 v(t) dt.$$

$$= -\int_0^3 (2t - 6) dt + \int_3^4 (2t - 6) dt.$$

$$= -\left(t^2 - 6t\right) \Big|_0^3 + \left(t^2 - 6t\right) \Big|_3^4$$

$$\begin{aligned}
 &= -(9 - 18) + (16 - 24 - (9 - 18)) \\
 &= -2(-9) - 8 \\
 &= 10.
 \end{aligned}$$

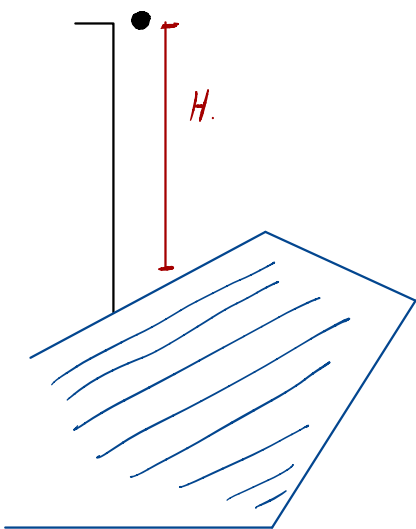
2nd way:



$$\begin{aligned}
 \mathcal{I}_d &= \int_0^4 |v(t)| dt = A_1 + A_2 \\
 &= \frac{1}{2} (3)(6) + \frac{1}{2} (1)(2) \\
 &= 9 + 1 = 10.
 \end{aligned}$$

Problem 3. A rock is dropped from a cliff and hits the surface of the lake below 3 seconds later. Find the distance between the rock and the surface of the lake as a function of the time starting from the moment it was dropped.

Note: Assume that the acceleration of gravity is 10 m/s^2 and ignore the drag.



$$a(t) = -10 \text{ m/s}^2.$$

$$v(t) = \int a(t) dt = \int (-10) dt = -10t + C.$$

$$\text{at } t=0 \text{ then } v(0) = 0 \text{ m/s}.$$

$$0 = -10(0) + C \quad \text{so } C = 0 \text{ m/s}.$$

$$\begin{aligned}
 \underbrace{[0]}_{\text{m/s}} &= \underbrace{[-10]}_{\text{m/s}^2} \times \underbrace{[0]}_{\text{s}} + \underbrace{[C]}_{\text{m/s}}.
 \end{aligned}$$

$$v(t) = -10t$$

$$H(t) = \int v(t) dt = \int -10t dt = -5t^2 + D.$$

when $t=38$ then $H(38)=0$. So,

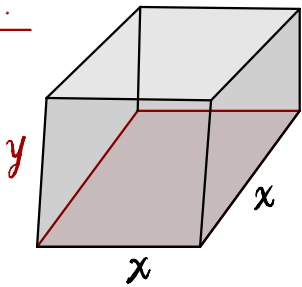
$$0 = -5(3)^2 + D$$

$$D = 45.$$

$$\text{So, } H(t) = -5t^2 + 45.$$

Problem 4. We are planning to make a box with a open top and square base. It has to have the volume of 6 ft^3 . The material for the base costs $\$3.00$ per square foot and the material for the sides costs $\$2.00$ per square foot. Find the dimensions of such box that result in the smallest possible total cost of material. justify.

soln.



$$V = x \cdot x \cdot y = x^2 \cdot y = 6 \text{ ft}^3$$

$$C = 3 \cdot x^2 + 2 \cdot (4xy)$$

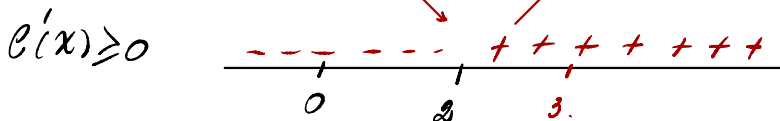
Cost base

$$\text{From } x^2 \cdot y = 6 \quad y = \frac{6}{x^2}$$

$$C(x) = 3x^2 + 8x \cdot \left(\frac{6}{x^2}\right) = 3x^2 + \frac{48}{x}$$

$$C'(x) = 6x - \frac{48}{x^2} = \frac{6x^3 - 48}{x^2} \Rightarrow \begin{matrix} x=0 \\ \times \end{matrix} \text{ or } x = \sqrt[3]{\frac{48}{6}} = 2.$$

The only critical point $x=2$.



By the 1st derivative test at $x=2$ we have a minimum.

$$y = \frac{6}{x^2} = \frac{6}{4} = \frac{3}{2}.$$

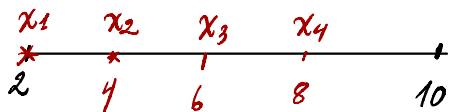
$2 \text{ ft} \times 2 \text{ ft} \times \frac{3}{2} \text{ ft}$ dimensions.

Problem 5. a). Calculate the left-point approximation L_n to integral.

$$\int_2^{10} \frac{1}{x} dx.$$

b). Draw the corresponding rectangles, together with graph of $f(x) = \frac{1}{x}$.

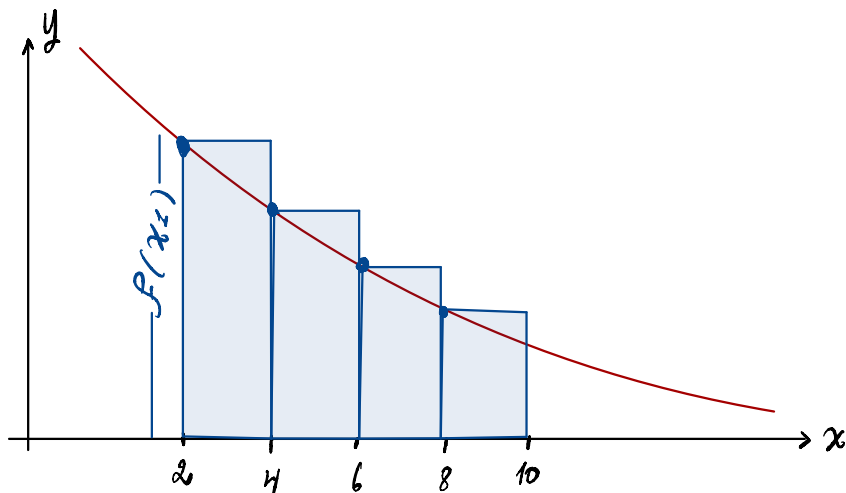
Sln. a).



$$\Delta x = \frac{10-2}{4} = 2.$$

$$\begin{aligned} L_4 &= f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x + f(x_4)\Delta x \\ &= 2(f(x_1) + f(x_2) + f(x_3) + f(x_4)) \\ &= 2\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8}\right) = \frac{25}{12}. \end{aligned}$$

b).



c). Is L_4 greater than the actual integral or less than it?

Specify and explain, or explain why this cannot be determined.

Sln c). L_4 is greater than $\int_2^{10} \frac{1}{x} dx$, b/c the total area of rectangles from part b). is greater than the area below the curve $y = f(x)$.

Problem 6. Express the following limit as an integral.

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \left\{ \sqrt{\left(\frac{1+8i}{n}\right)^2 + 3\left(\frac{1+8i}{n}\right)} \left(\frac{8}{n}\right) \right\}.$$

Sln.

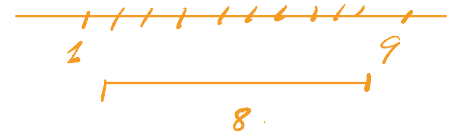
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \quad \Delta x = \frac{b-a}{n}.$$

$$x_i = a + i\Delta x.$$

$$\Delta x = \frac{8}{n}$$

$$x_i = 1 + \frac{8i}{n}$$

$$a=1, \quad b=9.$$



$$\int_1^9 \sqrt{x^2 + 3x} \, dx.$$

$$f(x) = \sqrt{x^2 + 3x}$$